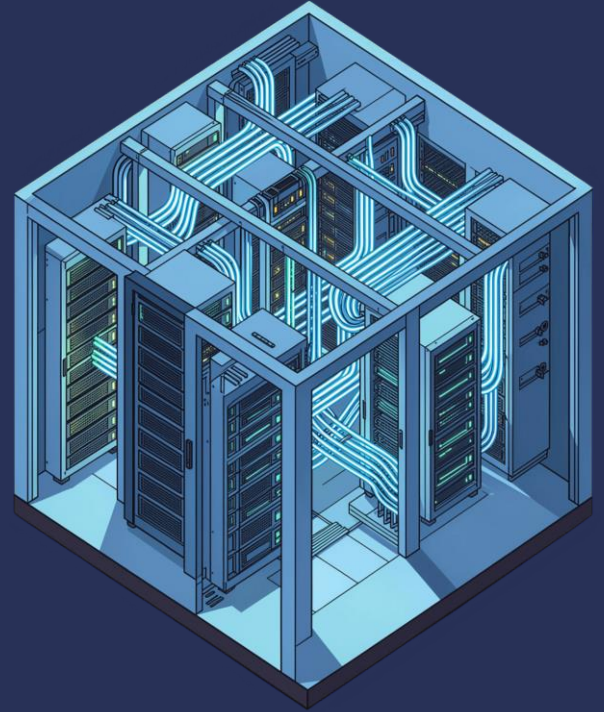


Network Monitoring Setup for Apex Corp

A comprehensive approach to network visibility, performance, and security.

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Phase 1 — Network Topology Setup

Built the Packet Tracer topology with Cisco 2911 (core router), Cisco 3560 (distribution switch), a Management Server, and two client PCs. All devices connected using copper straight-through cables. Inspected physical links and confirmed interface link status on each device.

Devices

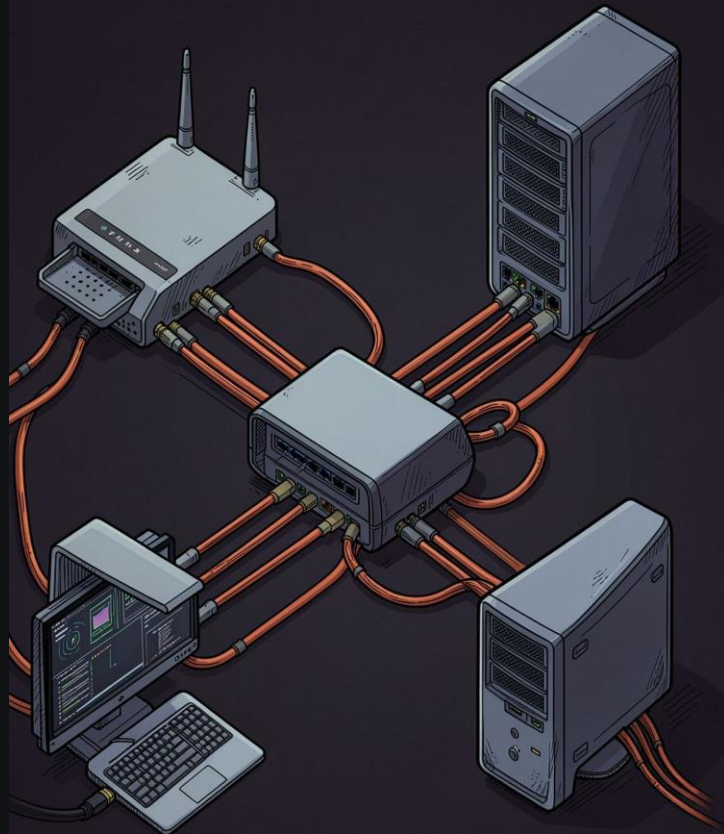
Cisco 2911, Cisco 3560,
Management Server, 2x PCs

Cabling

Copper straight-through:
router↔switch, switch↔PCs,
switch↔server

Verification

Checked link lights & interface status (show interfaces)





Phase 2 — IP Address Configuration

Assigned static IPv4 addresses across the topology using the 192.168.10.0/24 network. Configured subnet masks and appropriate default gateways on all devices to enable layer-3 communication and prepare for routing tests.

Addressing Plan

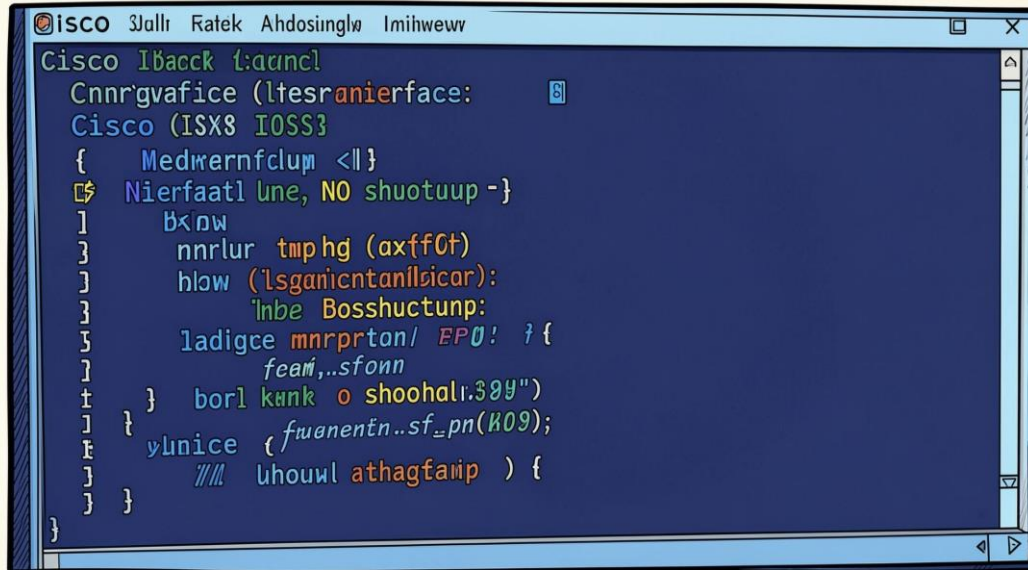
Network: 192.168.10.0/24 —
Router: .1, Server: .10, PC1:
.101, PC2: .102

Configuration Notes

Set subnet mask
255.255.255.0 and default-
gateway on hosts;
documented assignments

Phase 3 — Core Router Configuration

Configured GigabitEthernet0/0 on the Cisco 2911 with the assigned IP address and enabled the interface (no shutdown). Completed basic IOS hardening steps and validated operational status (show ip interface brief → Up/Up).



```

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```

Key Commands Used

- configure terminal
- interface GigabitEthernet0/0
- ip address 192.168.10.1 255.255.255.0
- no shutdown
- show ip interface brief

Phase 4 — Distribution Switch Configuration

Configured the Cisco 3560 management interface on VLAN 1 with a static management IP and assigned the default gateway to point at the router. Verified switch management reachability from the Management Server and a workstation.

- **Management Interface**

interface VLAN 1 → ip address 192.168.10.2 255.255.255.0

- **Gateway**

ip default-gateway 192.168.10.1

- **Validation**

ping from Management Server; tested SSH access once enabled



Phase 5 — Network Connectivity Testing

Performed end-to-end connectivity tests across the topology. Used ping and traceroute to validate reachability and pathing. Confirmed successful communication and corrected any IP or gateway misconfigurations discovered.

Test Procedures

- ping router from PCs and server
- ping between PCs
- traceroute to confirm hop path
- show ip route / show arp for verification



Phase 6 — Secure Remote Access (SSH)

Implemented SSH v2 on both router and switch to secure remote management. Created a local administrator account, generated RSA keys, disabled Telnet, and restricted VTY access to SSH.



Authentication

```
username admin privilege 15 secret [REDACTED]
```



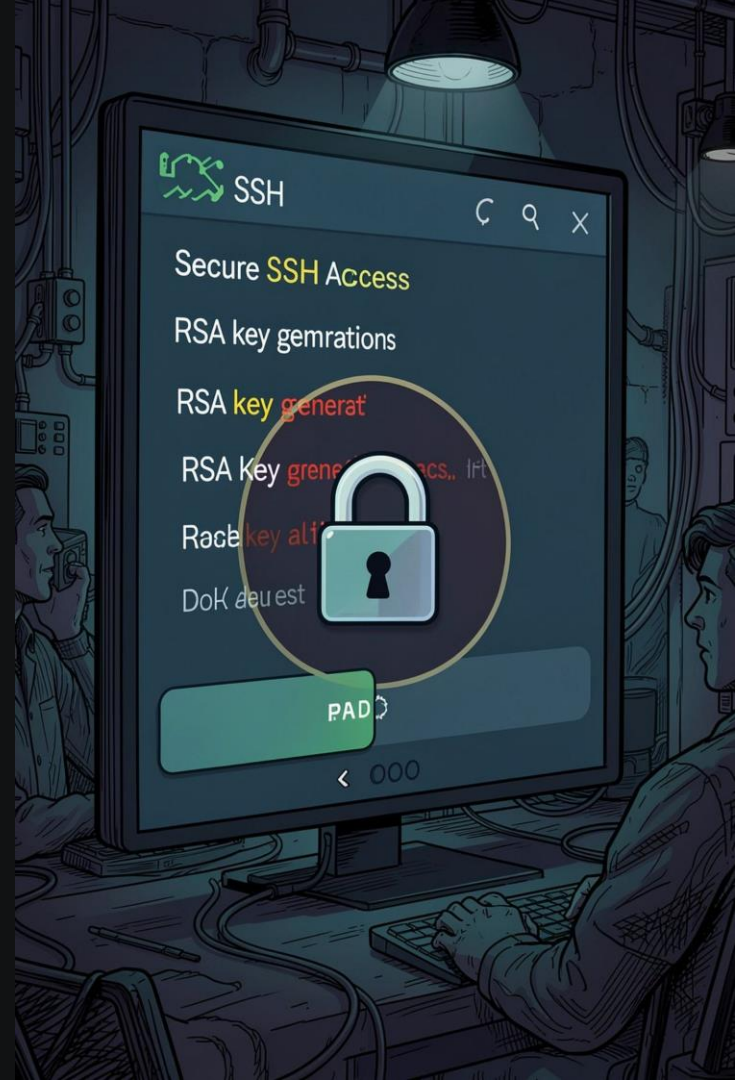
Encryption

```
crypto key generate rsa modulus 2048; ip ssh version 2
```



VTY Hardening

```
line vty 0 4 → transport input ssh; login local
```



Phase 7 — Syslog Configuration

Configured centralized syslog forwarding from router and switch to the Management Server. Enabled logging for interface up/down and configuration changes. Verified log receipt and timestamping on the server.



Configuration Summary

- logging host 192.168.10.10
- logging trap informational
- service timestamps log datetime msec
- verified logs with show logging and on-server viewer

Phase 8 — SNMP Configuration

Enabled SNMP read-only monitoring using community string 'public' for the lab environment. Registered devices with the Management Server and validated SNMP queries. Noted Packet Tracer limitations when interpreting SNMP traps and advanced MIB behavior.



SNMP Settings

```
snmp-server community public RO; snmp-server host 192.168.10.10 version 2c public
```



Packet Tracer Notes

Traps and some MIBs are simulated; production testing requires real devices

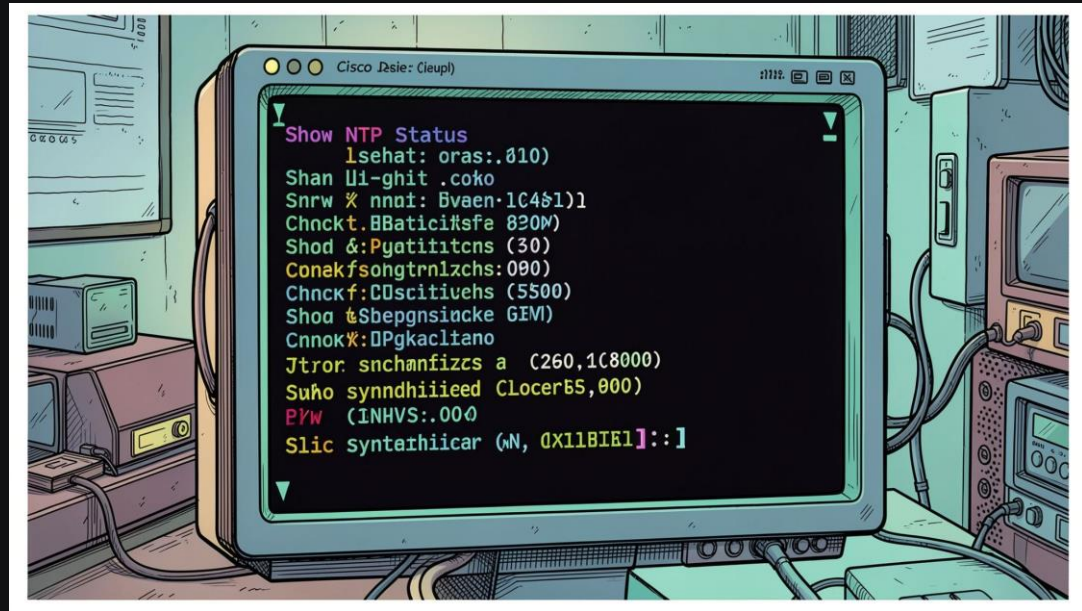


Phase 9 — NTP Configuration

Enabled NTP on the Management Server and attempted synchronization from the router and switch. Verified device clocks and recorded discrepancies. Documented Packet Tracer limitations for NTP accuracy and reachability in simulated environments.

Commands & Checks

- `ntp server 192.168.10.10`
- `show ntp status / show clock`
- ensure UDP/123 reachability between devices



Phase 10 — Configuration Change Monitoring

Implemented monitoring and verification of configuration changes using IOS commands. Added an MOTD security banner and confirmed change logging to syslog. Performed routine audits to ensure configuration integrity.

01

Audit Commands

show running-config; show startup-config;
show archive

03

Security Banner

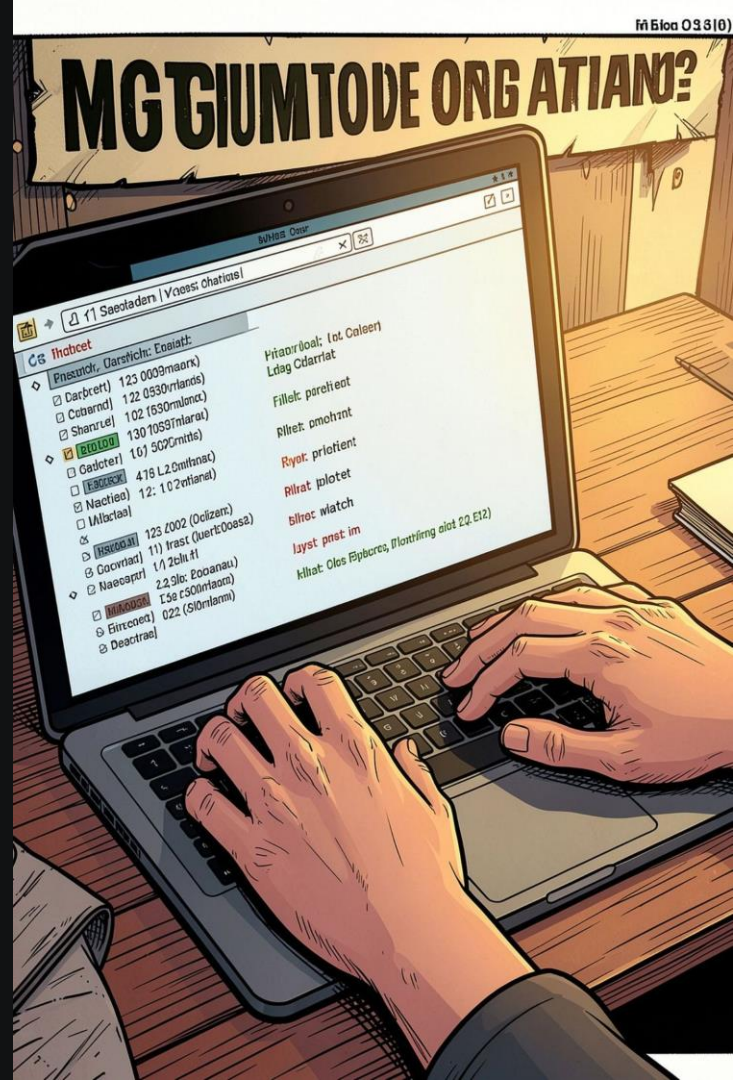
banner motd #Authorized personnel only. Activity monitored.#

- ❏ Routine verification and documented procedures reduce misconfiguration risk and support incident investigations.

02

Change Logging

enabled logging of configuration events to
syslog; timestamps included



Fixing Active Performance Issues

Addressing the current challenges in network performance is critical to maintaining efficiency and security.

1 Identify Performance Bottlenecks

Recognize areas where network speeds are lagging or resources are being overutilized, leading to slowdowns.

2 Analyze Traffic Patterns

Examine data flow and usage trends to pinpoint unusual spikes or drops in activity that may indicate issues.

3 Optimize Configurations

Adjust device settings and network configurations to improve performance and reduce latency.

4 Monitor Results

Continuously track network performance metrics to assess the effectiveness of implemented changes and ensure stability.



Conclusion and Next Steps

The implementation of a comprehensive monitoring setup will directly address the identified issues of intermittent slowdowns, unauthorized access attempts, and configuration changes. Continuous monitoring and proactive adjustments will ensure the network remains secure and performs optimally.

Slowdowns Resolved

Proactive performance monitoring eliminates recurring bottlenecks.

Security Strengthened

Real-time alerts guard against unauthorized access attempts.

Changes Tracked

Full configuration visibility ensures accountability and control.